**Oluwatosin Ayeni**

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16 October 2026

**Doubleword, London, UK**

Dear Hiring Team,

I am writing to apply for the Member of Technical Staff: Inference Research role at Doubleword. Your focus on multi-LoRA serving, GPU architecture, and speculative decoding is exactly the class of problem I have been building toward — and I have the benchmarks to prove it.

Three specific achievements relevant to this role:

- **H100 NVL inference research — RadixAttention and prefix caching:** I benchmarked vLLM v0.30 vs SGLang v0.5.20 on 2× NVIDIA H100 NVL 94 GB (NVLink 4.0, CUDA 13.4), Qwen3-8B BF16, TP=2. The critical finding was the impact of RadixAttention prefix caching in memory-constrained long-context settings: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context, with 100% SLO compliance at concurrency 128 and peak throughput of 5,965 tok/s. Results: inference-web-theta.vercel.app.
- **CUDA kernel profiling and speculative decoding:** Using NVIDIA Nsight Systems and Compute, I profiled KV cache CUDA ops on RTX 4090 hardware — identified memory bandwidth as the bottleneck and retuned kernel access patterns for 3.8× throughput improvement. I also ran speculative decoding experiments across vLLM and SGLang to characterise speedup vs draft model overhead tradeoffs, documented in *open-model-engine*.
- **Multi-modal LLM inference — vLLM-Omni:** I built *lab-vllm-omni*, a phased inference lab covering all 5 output modalities (text, audio, image, video, structured output) — characterising throughput, memory pressure, and scheduling behaviour across modalities. Directly relevant to serving diverse inference workloads efficiently.

Doubleword's approach — small team, deep technical work, research-to-production focus — is exactly what I want. I am based in Glasgow and happy to relocate to London; Farringdon 2–3 days/week is workable.

Yours sincerely,

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